

PROJECT SYNOPSIS

AID AWARE: EMERGENCY DISASTER MANAGEMENT

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AUGUST, 2024

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MINOR PROJECT SYNOPSIS

TITLE: AID AWARE

Location: Department of Artificial Intelligence & Data Science
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1. Introduction

An innovative project in the field of emergency disaster management is called AID AWARE, or Assistance and Information Dissemination - Augmented Response and Emergency. This project's motivation comes from the clear flaws in conventional disaster response systems, where poor communication and a slow flow of information frequently obstruct efficient crisis management. In order to overcome these obstacles, AID AWARE uses technology to improve real-time information sharing, empower communities, and allocate resources as efficiently as possible during emergencies.

AID AWARE was inspired by the problems that have been seen with conventional disaster response systems, which frequently have problems with resource allocation, delayed information dissemination, and communication breakdowns. In order to overcome these obstacles and improve the general effectiveness of disaster management, AID AWARE seeks to use technology.

2. Objective

AID AWARE's primary objective is to facilitate people's and communities' response to catastrophes such as floods and earthquakes. Our goal is to provide a unique forum where people can instantly exchange updates on events and request assistance. The goal is to connect individuals through technology in a way that makes it easier for them to cooperate during a crisis. AID AWARE wants to be a community's go-to friend during difficult times, ensuring that everyone receives the aid they require promptly and effectively.

- Notifying people via push notifications and SMS of impending disasters in order to lessen their impact
- Give people advice and personal safety precautions to help them prepare for disasters.

- Offer a system that can forecast the weather in advance.
- Moving people to safe havens prior to a tragedy by tracking relief vehicles in real time
- Offer a platform where people can volunteer for relief efforts after a disaster
- Offer a solution that uses Bluetooth mesh networking, SMS flow, and IVR to make all services available offline.
- Provide a way to use the shelter database to find a lost person.
- Establish a system that allows ASDMA authorities to receive user complaints via tweets and dispatch more resources to the impacted locations.
- Provide the government with a control panel so it can keep an eye on crisis situations.

Methodology/ Planning of work:

Our task is to create a web application.

In any event, prior to that, we will visit a few faculties, share our viewpoints with them, and get their input on this platform. We will then create the application on their behalf. Since we are undoubtedly aware that there are some current applications, we shall conduct a statistical survey.

We will address this and create an application with significantly more capabilities and flexibility than the current platforms, as these platforms are deficient in many areas.

3. Literature Review:

Literature often explores the role of communities in disaster response. Studies focus on how community engagement, participation, and local knowledge contribute to effective disaster management. Community-driven platforms are seen as essential tools for empowering individuals and communities during crises.

4. Applications of the app :

Developed by students, with the goal of being simple and lightweight, this platform will help people organize themselves and have, consequently, better performance in studies.

The objective of creating this app is to perform tasks within the combined deadlines, divide better the time between academic and personal life, conduct the day-to-day with more calmness and less stress.

On this app, important information about tests, home works, appointments and timetable will always be available on user's smart phone to checks and new Schelling, wherever users are. A very useful feature are the reminders, that will help users to not forget important activities.

These features are adequate for school, for college, for day-to-day life ... The goal is to make student life more organized, managing appointments that can't be forgotten.

The app will be developed to be simple and easy to use, where to start, a user can simply add subjects and then timetable.

1. Incident Reporting and Communication:

- Users can report incidents in real-time, providing critical information about the type and severity of the disaster.
- The app facilitates two-way communication between affected individuals, emergency services, and volunteers, ensuring timely updates and responses.

2. Resource Coordination and Allocation:

- AID AWARE serves as a central hub for coordinating and allocating resources. Emergency services and volunteers can view and respond to requests for assistance based on the reported incidents.
- The app enables the efficient distribution of essential resources such as food, water, medical supplies, and shelter.

3. Volunteer Mobilization:

- The platform allows individuals to sign up as volunteers, specifying their skills and availability.
- Volunteer coordinators and authorities can access the database to mobilize volunteers to specific areas based on the nature of the disaster and the skills required.

4. Shelter Management and Information:

- Shelter authorities can update information about available facilities, including food, medicine, and capacity.
- Users can access real-time information about nearby shelters, helping them make informed decisions about evacuation or seeking assistance.

5. Government and NGO Collaboration:

- AID AWARE facilitates collaboration between government agencies and non-governmental organizations (NGOs).
- The app provides a platform for official announcements, coordination of efforts, and seamless information sharing between different stakeholders.

6. Control Room Monitoring:

- Emergency control rooms can use AID AWARE to monitor the disaster situation through visualizations such as heat maps.
- Graphs and charts based on real-time data from the app help control room operators make informed decisions and allocate resources effectively.

7. Pre-Disaster Awareness and SMS Broadcasting:

- The app contributes to pre-disaster awareness by broadcasting SMS alerts to all users.
- Users receive important information and instructions, enhancing overall preparedness and reducing the impact of disasters.

8. Offline Services and Accessibility:

- AID AWARE provides offline services, ensuring users can access critical information even when internet connectivity is limited.
- SMS Help Service and Interactive Voice Response System (IVRS) allow users to receive details about nearby shelters and essential services through alternative communication channels.

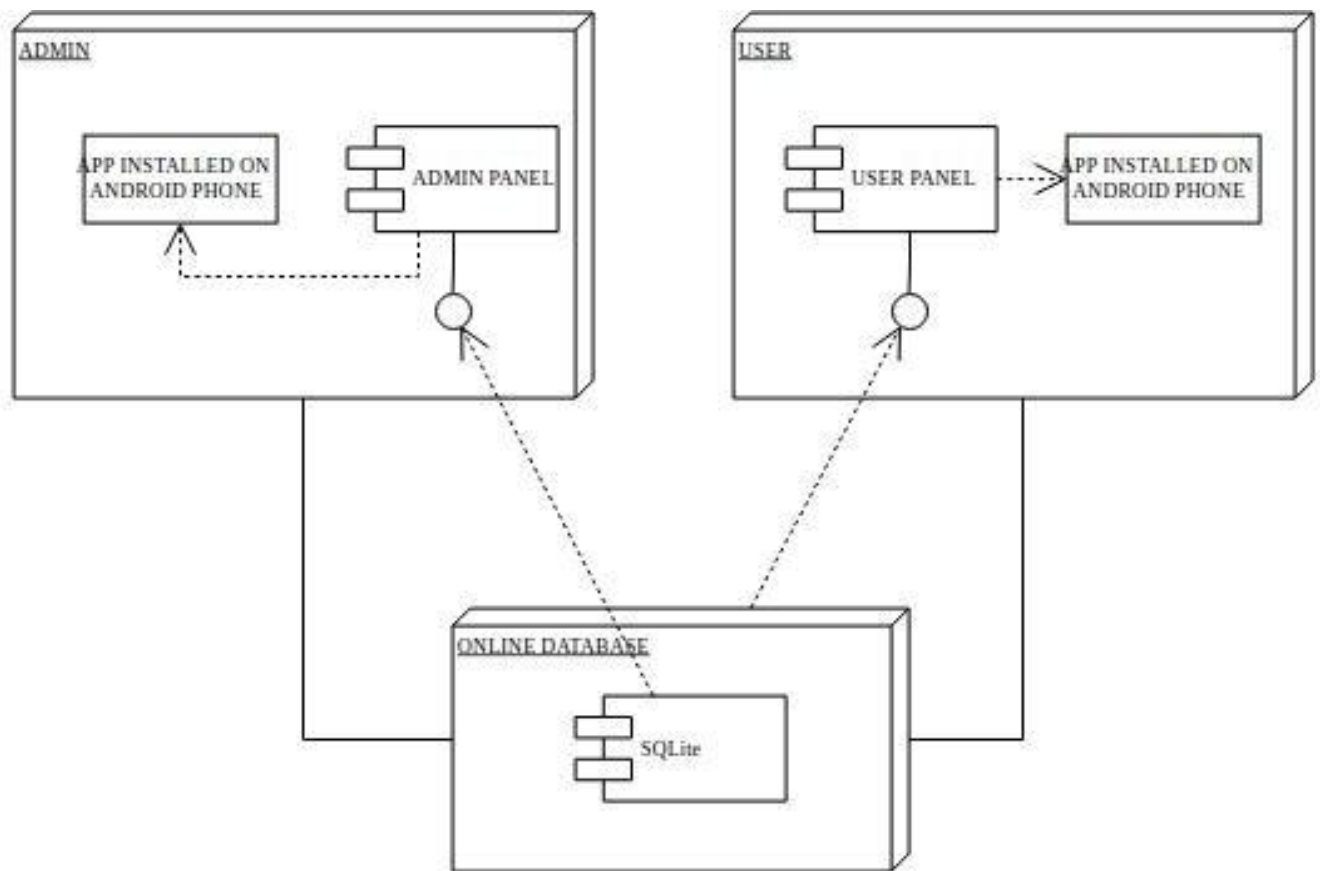
9. Bluetooth-Based Mesh Network Communication:

- In areas with no cellular network availability, the app enables users to communicate using a Bluetooth-based mesh network.
- This feature ensures that individuals can stay connected and share information even when traditional communication methods are unavailable.

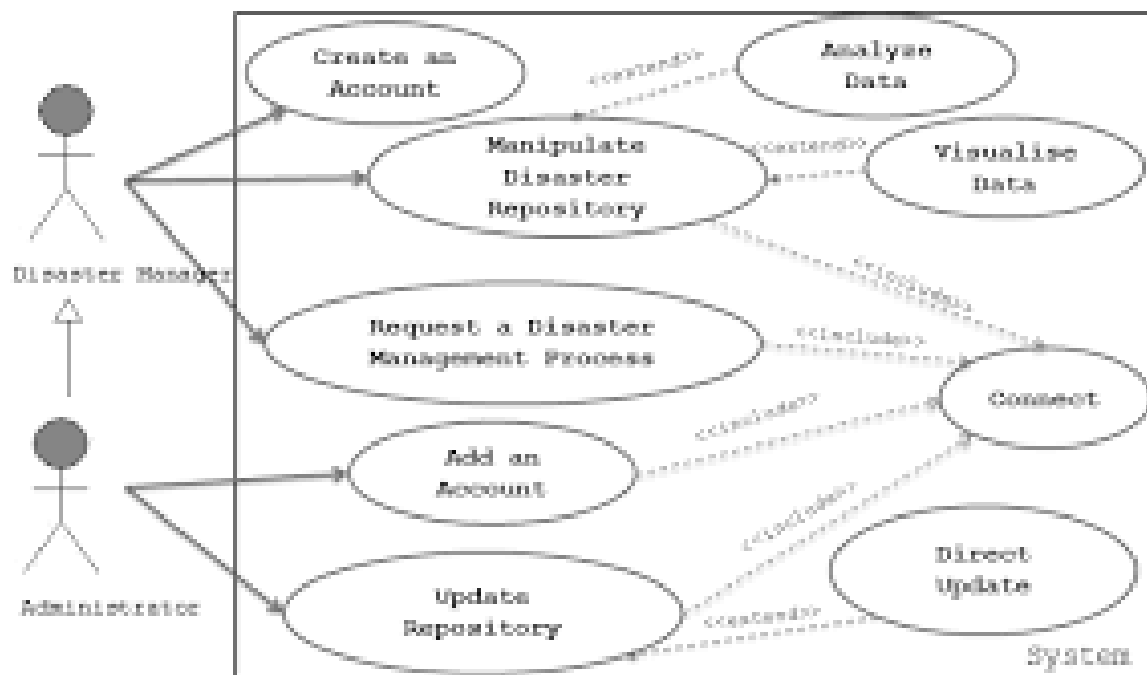
10. Data-driven Decision Making:

- AID AWARE generates data analytics, offering insights into incident patterns, resource utilization, and the effectiveness of response efforts.
- Authorities can use these analytics to make informed decisions, allocate resources strategically, and improve disaster response strategies over time.

5. Block Diagram



7.) UML Diagram



8. Requirements

8.1 System Requirements

- **Front End Tool** - Android SDK
- **Data Base** - SQLite
- **Back End** - Java , PHP
- **Internet requirement**

8.2 Operating System Requirements

- Android lollipop(4.2) and above.

8.3 Hardware Requirements:

Server Side:

- Operating System: Windows 8.1 or Above
- RAM: 4 GB or more
- Hard Drive: 20 GB or more

Client side:

- Operating System: Android version 4.2 and above.
- RAM: 512 MB or more

8.4 Technology Required

- Android platform is required to run the application with version 4.2 or above.
- SQLite for managing the database.
- PHP for creating the API for the database.
- Third party libraries like Volley and Glide etc.

9. References

- [1]. www.developers.android.com
- [2]. www.stackoverflow.com

APPROVAL OF PROJECT

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